

SEORIM PARK



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Crissier, Switzerland

South Korean

TECHNICAL SKILLS:

- Programming languages:

- Python:

- Data manipulation and visualization: **pandas, seaborn, matplotlib, numpy**

- Generation of interactive plots: **plotly, dash, bokeh**

- Machine learning framework: **flax, scikit-learn, tensorflow**

- R:

- Data manipulation and visualization: **dplyr, ggplot2**
- Interactive plots: **Shiny**

- Biostatistics tools: **bioconductor**

- C, C++, MATLAB:

- Implementation of mathematical models and algorithms

LANGUAGE (CEFR):

- **Korean:** Native

- **English:** C1

- **French:** C2

PROFILE

Master's graduate in Applied Mathematics from EPFL, with strong foundations in mathematics, data analysis and machine learning. Experienced in interdisciplinary research, through projects in biostatistics, computational biology and deep learning.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

- **Master of Science in Applied Mathematics**, Sept. 2021 - Feb. 2025
overall grade 5.15/6

École Polytechnique, Palaiseau, France

Sept. 2018 - July 2021

- **Bachelor of Science, graduated Cum Laude with a major in Mathematics & Computer Science and a minor in Biology**

WORK EXPERIENCE

Master's Thesis at the Chair of Dynamics of Learning Algorithms, EPFL

Sept. 2024 - Feb. 2025

- **Thesis supervised by Lenaïc Chizat and Pierre Marion:** Explored mathematical foundations of **recent deep learning methods based on Recurrent Neural Networks and Transformers**, and measured the trade-off between performance and efficiency of each model.
- Conducted a comparative study on **Linear Recurrent Units** to observe the impact of hyperparameter tuning and model architecture on their performance.

Internship at Nestlé Institute of Health Sciences (NIHS), Lausanne

June 2023 - Dec. 2023

- **Analyzed large-scale longitudinal multi-omics datasets** with statistical tools such as MEFISTO, Non-negative Tensor factorization (TCAM) or Functional PCA on longitudinal samples from cancer patients (metabolomes, proteomes and immune markers), microbiome samples from participants with diet intervention, or metabolomic data from pets with a target disease.
- Collaborated with experts from diverse fields to analyze multi-omics samples from breast milk and fecal samples from infants to measure the nutritional impact of a dairy product developed at Nestlé.

PROJECTS

- **Machine learning:** Engaged in multiple projects such as the comparative study of classifiers on label propagation for semi-supervised learning and the experimental study on the edge of stability of gradient descent in neural networks.
- **Statistics:** Collaborated on several projects involving statistical modeling and interactive visualization of real-world data ([link GitHub](#)).
- **Biostatistics:** Experience in the application of several biostatistical methods such as Genome-Wide Association studies or the analysis of Affymetrix RNA microarrays samples. Contributed to the construction of DNA analysis pipeline in the interdisciplinary association Genorobotics at EPFL.
- **Mathematics and Computer Science:** Conducted a semester-long research project exploring network summary plots and implemented the method in python to test on real-world graph data ([link GitHub](#)). Implemented the Shortest Path algorithm from a given literature in C on graph data with parallel computation ([link GitHub](#)).